

Volunteer Hours Management API

API Documentation

COMP3011 Web Services and Web Data | Coursework 1

Live deployment: <https://volunteer-hours-api-cw1.onrender.com>

1. Overview

The Volunteer Hours Management API is a FastAPI backend for managing volunteers, events, shifts, work logs, CSV imports, analytics, and project statistics. It is designed for event organisers who need to track volunteer participation accurately and generate useful summaries from stored data.

2. Live Deployment

Page	URL
Homepage	https://volunteer-hours-api-cw1.onrender.com/
API Docs	https://volunteer-hours-api-cw1.onrender.com/docs
Admin Dashboard	https://volunteer-hours-api-cw1.onrender.com/dashboard
Health Check	https://volunteer-hours-api-cw1.onrender.com/health

3. Authentication

Most routes are protected and require a valid JWT access token in the Authorization header.

Auth flow:

1. Register with POST /auth/register
2. Login with POST /auth/login
3. Copy the returned access_token
4. In Swagger /docs, click Authorise and paste: Bearer YOUR_ACCESS_TOKEN

Roles: admin | organiser

Admin-only routes: all /imports/* endpoints and volunteer deletion.

4. Route Groups

- `/` - Homepage with accessibility toolbar and navigation
- `/dashboard` - Admin statistics dashboard (JWT-authenticated via browser)
- `/auth` - Registration, login, current user
- `/volunteers` - Volunteer CRUD and hours summary
- `/events` - Event CRUD and hours summary
- `/shifts` - Shift CRUD nested under events
- `/work-logs` - Work log CRUD with automatic duration calculation
- `/imports` - Admin-only CSV import endpoints
- `/analytics` - Leaderboard, awards, volunteer summaries
- `/stats` - System-wide totals (authenticated)
- `/health` - Service health check

5. Endpoints

5.1 Home and Dashboard

GET

`/` - Homepage

Returns the project homepage with feature cards, navigation links, text size controls, and a high-contrast greyscale colour-blind mode (WCAG 2.1 Level AAA, 21:1 contrast ratio).

GET

`/dashboard` - Admin Statistics Dashboard

Returns an interactive HTML page where an admin pastes a JWT token to view live system totals (volunteers, events, shifts, work logs, worked minutes, worked hours) displayed as animated stat cards.

5.2 Health

GET

`/health` - Health check

```
{ "status":  
  
  "ok" }
```

5.3 Authentication

POST

`/auth/register` - Register a new user

Request:

```
{  
  "email": "admin@test.com",
```

```
{
  "full_name": "Admin User",
  "password": "Test12345!",
  "role": "admin"
}
```

Response:

```
{
  "id": 1,
  "email": "admin@test.com",
  "full_name": "Admin User",
  "role": "admin"
}
```

POST

/auth/login - Login and receive JWT token

Request:

```
{
  "email": "admin@test.com",
  "password": "Test12345!"
}
```

Response:

```
{
  "access_token": "your.jwt.token",
  "token_type": "bearer"
}
```

GET

/auth/me - Get current authenticated user

```
{
  "id": 1,
  "email": "admin@test.com",
  "full_name": "Admin User",
  "role": "admin"
}
```

5.4 Volunteers

POST

/volunteers - Create volunteer

```
{
  "volunteer_no": "V001",
  "name": "Ahmed Ali",
  "email": "ahmed@test.com",
  "phone": "772100000"
}
```

GET	/volunteers - List all volunteers
GET	/volunteers/{volunteer_id} - Get volunteer by ID
PATCH	/volunteers/{volunteer_id} - Update volunteer

```
( "phone": "770000001" )
```

DELETE	/volunteers/{volunteer_id} - Delete volunteer
GET	/volunteers/{volunteer_id}/hours - Get volunteer total hours

```
{
  "target_id": 1,
  "worked_minutes": 150,
  "worked_hours": 2.5
}
```

5.5 Events

POST	/events - Create event
------	------------------------

```
{
  "title": "Saudi National Day",
  "description": "Main event",
  "location": "Leeds",
  "event_date": "2025-09-27"
}
```

GET	/events - List all events
GET	/events/{event_id} - Get event by ID
PATCH	/events/{event_id} - Update event
DELETE	/events/{event_id} - Delete event
GET	/events/{event_id}/hours - Get event total volunteer hours

```
{
  "target_id": 1,
  "worked_minutes": 480,
  "worked_hours": 8.0
}
```

5.6 Shifts

<pre>{ "name": "Morning Shift", "description": "Setup and preparation", "start_time": "2025-09-27T09:00:00", "end_time": "2025-09-27T12:00:00" }</pre>	
GET	/events/{event_id}/shifts - Create shift for event
GET	/shifts/{shift_id} - List shifts for event
GET	/shifts/{shift_id} - Get shift by ID
PATCH	/shifts/{shift_id} - Update shift
DELETE	/shifts/{shift_id} - Delete shift

5.7 Work Logs

<pre>{ "volunteer_id": 1, "shift_id": 1, "checked_in_at": "2025-09-27T09:15:00", "checked_out_at": "2025-09-27T11:45:00" }</pre>	
Response:	
<pre>{ "id": 1, "volunteer_id": 1, "shift_id": 1, "checked_in_at": "2025-09-27T09:15:00", "checked_out_at": "2025-09-27T11:45:00", "worked_minutes": 150 }</pre>	
GET	/work-logs/{work_log_id} - Create work log
GET	/work-logs/{work_log_id} - Get work log by ID
PATCH	/work-logs/{work_log_id} - Update work log
DELETE	/work-logs/{work_log_id} - Delete work log

Worked-hours calculation logic:

```
effective_start = max(checked_in_at, shift.start_time)
effective_end   = min(checked_out_at, shift.end_time)
worked_minutes  = max(0, effective_end - effective_start)
```

Validation rejects: checkout before check-in, duplicate logs for the same volunteer and shift.

5.8 Imports (Admin only)

POST	/imports/volunteers - Upload volunteer CSV
POST	/imports/events - Upload events CSV
POST	/imports/attendance - Upload attendance CSV

All import endpoints accept multipart/form-data file uploads and require admin authentication.

The import system normalises column names at ingestion time, supports alternate headers, and handles partial data gracefully (missing email/phone are allowed for volunteers).

Example response:

```
{
  "created": 10,
  "updated": 2,
  "skipped": 1
}
```

5.9 Analytics

GET	/analytics/leaderboard - Volunteers ranked by total hours
-----	---

```
[
  {
    "volunteer_id": 1,
    "name": "Ahmed Ali",
    "total_minutes": 600,
    "total_hours": 10.0
  }
]
```

GET	/analytics/awards - Volunteers grouped into award tiers
-----	---

Tiers: tier_a (20+ hrs) | tier_b (15–20 hrs) | tier_c (1–15 hrs)

```
{
  "tier_a": [],
  "tier_b": [],
  "tier_c": [
    { "volunteer_id": 1, "name": "Ahmed Ali", "total_minutes": 600,
      "total_hours": 10.0 }
  ]
}
```

```
]
}
```

GET

/analytics/volunteers/{volunteer_id}/summary - Comprehensive volunteer summary

```
{
  "volunteer_id": 1,
  "name": "Ahmed Ali",
  "total_events_worked": 2,
  "total_shifts_worked": 3,
  "total_minutes": 600,
  "total_hours": 10.0,
  "recent_activity": [
    {
      "event_id": 1,
      "event_title": "Saudi National Day",
      "shift_id": 2,
      "shift_name": "Evening Shift",
      "checked_in_at": "2025-09-27T17:00:00",
      "checked_out_at": "2025-09-27T19:00:00",
      "worked_minutes": 120,
      "worked_hours": 2.0
    }
  ]
}
```

5.10 Stats

GET

/stats — System-wide totals (authenticated)

Requires authentication. Use /dashboard to view in a browser without writing code.

```
{
  "total_volunteers": 116,
  "total_events": 108,
  "total_shifts": 100,
  "total_work_logs": 383,
  "total_worked_minutes": 78120,
  "total_worked_hours": 1302.0 }
```

6. Error Codes

Code	Meaning
200 OK	Successful read or update

201 Created	Successful resource creation
204 No Content	Successful deletion
400 Bad Request	Invalid request or business-rule violation
401 Unauthorized	Missing or invalid JWT token
403 Forbidden	Authenticated but not permitted for this action
404 Not Found	Requested resource does not exist
422 Unprocessable Entity	Request body validation error

Example validation error:

```
{  "detail": "Validation
failed",
  "errors": [
    {
      "loc": ["body", "email"],      "msg":
"value is not a valid email address",
      "type": "value_error"
    }
  ]
}
```

Example resource error:

```
{ "details": "Volunteer not found"}
```

7. Notes

The base path / returns a homepage - not a 404 response.

The admin dashboard at /dashboard provides a visual stats view using a JWT token passed client-side.

Auth registration uses full_name; volunteer payload uses name - this is intentional for schema compatibility.

Analytics are most meaningful after importing data or creating work logs manually.

Import endpoints are admin-protected and accept multipart/form-data file uploads.

Real datasets are included under datasets/ in the repository for testing and demonstration.